Signals and Systems — Tutorial 6

System response and stability

[Tutor version — includes solutions]

Tutorial information

- It is expected that you attempt the prework questions before coming to the tutorial each week and submit your *handwritten* solutions to your tutor within the first 15 minutes of your tutorial.
- Not all questions are equally difficult. Difficult questions are marked with a $\star$.
- Questions marked **Extension** go beyond the assessable content for this course and are provided to give a deeper theoretical background and understanding.
- You may not be able to complete all questions within the tutorial time. It is *strongly* recommended that you attempt all questions in your own time and ask your tutor if you are stuck.

Prework questions

A: Impulse Response

Suppose the LTI systems $y_r(t), y_b(t), y_p(t)$ and $y_g(t)$ have impulse responses $h_r(t), h_b(t), h_p(t)$ and $h_g(t)$ shown in Figure 1. Identify which of the systems are BIBO stable.

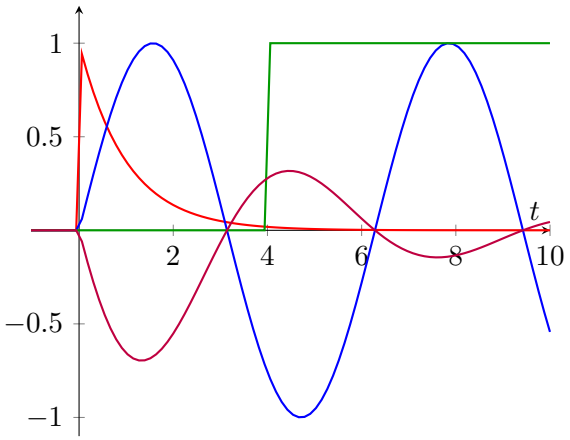


Figure 1: Impulse responses for the LTI systems y_r, y_b, y_p and y_g.

> A system is BIBO stable if and only if its impulse response $h(t)$ satisfies
>
> $$\int_{-\infty}^{\infty} |h(t)|\, dt < \infty$$

- y_r and y_p are BIBO stable as it appears that both the impulse responses h_r and h_p are converging to 0 and the integral is likely bounded.
- y_b appears to not decay so it is likely not BIBO stable.
- $y_g(t) = H(t-4)$ so it is not BIBO stable.

B: Stability

Let $\dot{y} = f(u(t), y(t))$ be a LTI system. Recall from Tutorial 1 the system has solutions of the form $y(t) = Ae^{zt} + B$ which we can plot in the complex plane.

Determine if the systems in the following plots are stable or unstable and if they are stable if they are critically/over-damped or under-damped.

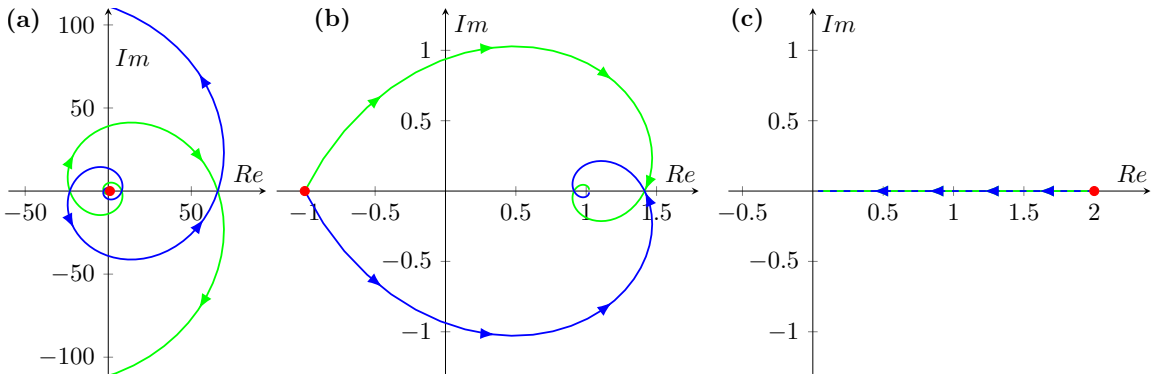


Figure 2: Plots of solutions $y(t) = Ae^{zt} + B$ where $A, B \in \mathbb{R}$, $z \in \mathbb{C}$ and $t \geq 0$. The blue line is when $z = x + jy$ and the green is when $z = x - jy$. The initial condition is shown by the red point.

a. The system is unstable as the trajectories diverge i.e. the magnitude of the solution $|y(t)| \to \infty$ as $t \to \infty$. As it is unstable it is neither over, critically nor underdamped.

b. The system is stable as the trajectories converge to the point $1+0j$. As the trajectories spiral and have non-zero imaginary component the system is underdamped. Notice that the trajectories are complex conjugates of each other so the sum of the two trajectories will have a sinusoidal term.

c. The system is stable as the trajectories converge to the origin. As both trajectories remain confined to the real axis the system is either critically or over damped at the solution is $y(t) = 2e^{-at}$ for some $a > 0$.

Tutorial questions

1: Stability

1. Determine if the following systems are stable or unstable. If stable, find the damping coefficient and natural frequency.

   (a) $\dot{x} + 2x = 4u(t)$

   (b) $4\ddot{x} - 2\dot{x} + (5/2)x = u(t)$

   (c) $\ddot{x} + 2\dot{x} + 5x = 5u(t)$

Recall from the lecture notes that a system defined by a differential equation is stable if and only if all its poles are in the LHP.

(a) $\dot{x} + 2x = 4u$ has transfer function

$$G(j\omega) = \frac{4}{j\omega + 2}$$

which has a stable pole at -2. However as the system is first order it does not have a well-defined damping ratio or natural frequency.

(b) $4\ddot{x} - 2\dot{x} + (5/2)x = u$ has a transfer function

$$G(j\omega) = \frac{2}{8(j\omega)^2 - 4j\omega + 5} = \frac{16}{5}\left(\frac{5/8}{(j\omega)^2 - (1/2)j\omega + 5/8}\right)$$

which has poles at $0.25 \pm 0.75j$ which are unstable. To find the natural frequency and damping ratio we match coeffecients with the standard second-order form

$$G(j\omega) = K\frac{\omega_n^2}{(j\omega)^2 + 2\zeta\omega_n j\omega + \omega_n^2}$$

we have $\omega_n = \sqrt{5/8}$ and $\zeta = -1/(4\omega_n) = -0.32$.

(c) $\ddot{x} + 2\dot{x} + 5x = 5u$ has a transfer function

$$G(j\omega) = \frac{5}{(j\omega)^2 + 2j\omega + 5}$$

which has poles at $-1 \pm 2j$ which are stable. Similarly to the above we have $\omega_n = \sqrt{5}$ and $\zeta = 1/\sqrt{5}$.

Similarly a system is stable if and only if its impulse response $h(t)$ satisfies

$$\|h(t)\|_1 = \int_{-\infty}^{\infty} |h(t)|\, dt < \infty$$

The impulse response of each system is

(a) $h(t) = 4e^{-2t}H(t)$. So $\|h(t)\|_1$ is bounded

(b) $h(t) = \frac{e^{t/4}}{3}\sin((3/4)t)\,H(t)$. So $\|h(t)\|_1$ is unbounded

(c) $h(t) = \frac{5e^{-t}}{2}\sin(2t)\,H(t)$. So $\|h(t)\|_1$ is bounded.

We can also use the definition

$$\max_t\{|y(t)|\} \leq \gamma \max_t\{|u(t)|\}$$

which holds for (a) and (c) by taking $\gamma = \|h(t)\|_1$.

2: System Identification and Bode Plots

Three systems $y_1(t), y_2(t)$ and $y_3(t)$ have Bode magnitude plots in Figure 3. Identify which of the impulse responses $h_r(t), h_b(t), h_p(t)$ and $h_g(t)$ correspond with each of the Bode plots.

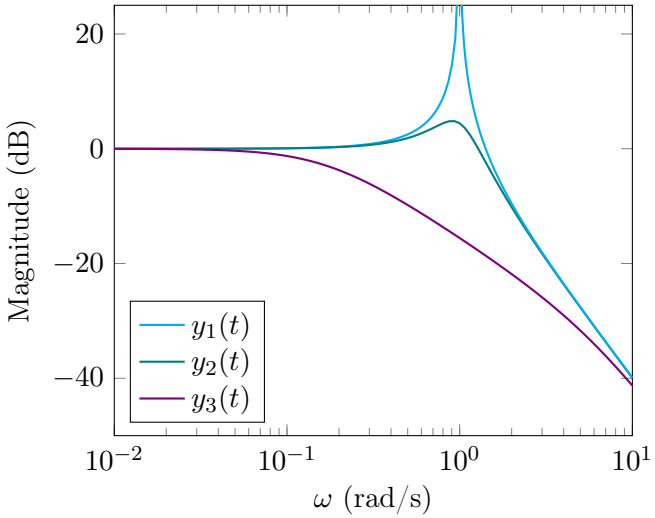


Figure 3: Bode magnitude plots for the systems y_1, y_2 and y_3

- The Bode magnitude plot y_3 for a critically or overdamped system so it corresponds to the red impulse $h_r(t)$.
- Both y_1 and y_2 are Bode magnitude plots for underdamped systems so could correspond with either of the impulse responses $h_b(t)$ or $h_p(t)$.
- As y_2 has a lower resonant peak (is more damped) than y_1, y_1 corresponds with h_b as the amplitude of the sinusoid does not appear to decay and
- y_2 corresponds with h_p as the sinusoid seems to decay.

3: Resonance and Type 1 Diabetes

Type one diabetics no longer produce endogenous insulin and rely on the exogenous delivery of insulin to manage and lower blood glucose levels. Typically insulin is delivered via an insulin pump in discrete boluses of size A_k at some frequency ω_0 i.e.

$$`u(t) = \sum_k A_k \delta(t - kT)$$

where T is the period. A group of type one diabetics are administered insulin at different frequencies over a twenty-four hour period and the relative increase or decrease of their blood glucose level is recorded. These data are plotted in Figure 4. The total amount of insulin delivered is constant.

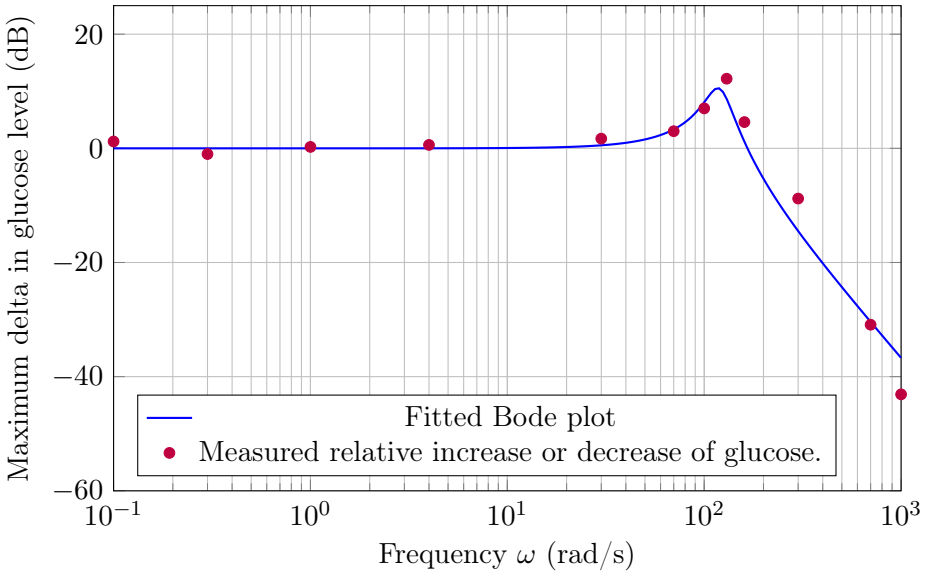


Figure 4: Data and fitted frequency response.

A transfer function

$$H(j\omega) = \frac{G(j\omega)}{U(j\omega)} = \frac{14.4 \times 10^3}{(j\omega)^2 + 36j\omega + 14.4 \times 10^3} \tag{1}$$

is fitted to the data.

1. Is the system BIBO stable? Justify your answer by finding the impulse response and poles of the system.
2. Find the natural frequency ω_n and damping ratio ζ for the system.
3. Find the ω_0 which maximises the effectiveness of insulin. What does this ω_0 represent?
4. Use the inverse Fourier transform to find a model for insulin-glucose dynamics in type one diabetics. How would we identify the optimal frequency to bolus insulin from this time domain model?
5. If this were a model for an RLC series circuit where the input is the voltage of the supply and the output is the voltage across the capacitor give values for the resistor, capacitor and inductor such that transfer function of the circuit matches (1).

1. The denominator of the transfer function $H(j\omega)$ is $(j\omega)^2+36j\omega+14.4\times10^3$ which means the poles of the system occur at $-18\pm6\sqrt{391}j$ which have negative real part so are stable.

   To find the impulse response of the system we find the inverse Fourier transform. Defining $\omega_d=\omega_n\sqrt{1-\zeta^2}$, we have

   $$G(j\omega)=\frac{\omega_n^2}{(j\omega)^2+2\zeta\omega_n j\omega+\omega_n^2}=\frac{\omega_n^2}{(j\omega+\zeta\omega_n)^2-\omega_d^2}=\left(\frac{\omega_n^2}{\omega_d}\right)\left(\frac{\omega_d}{(j\omega+\zeta\omega_n)^2+\omega_d^2}\right)$$

   Using the transform pair

   $$e^{-at}\sin(\omega_0 t)\,H(t)\leftrightarrow\frac{\omega_0}{(j\omega+a)^2+\omega_0^2}$$

   We have the impulse response is

   $$h(t)=\left(\frac{\omega_n^2}{\omega_d}\right)e^{-\zeta\omega_n t}\sin(\omega_d t)H(t)$$

2. We equate the transfer function with a transfer function in standard form

   $$H(j\omega)=\frac{\omega_n^2}{(j\omega)^2+2\zeta\omega_n(j\omega)+\omega_n^2}$$

   equating coeffecients we have

   $$2\zeta\omega_n=36\text{ and }\omega_n^2=14.4\times10^3$$

   which gives $\omega_n=120$ and $\zeta=0.15$.

3. Insulin is most effective when the gain $|H(j\omega)|$ is maximised. From inspection of Figure 4 this occurs at roughly $\omega\sim10^2$. To solve this rigorously we find the maximum of $|H(j\omega)|$. Setting $a=36$ and $b=14.4\times10^3$ we have

   $$\begin{aligned}\frac{d}{d\omega}|H(j\omega)|&=\frac{d}{d\omega}\left(\frac{b}{|(j\omega)^2+a(j\omega)+b|}\right)\\&=\frac{d}{d\omega}\left(\frac{b}{\sqrt{(b-\omega^2)^2+a^2\omega^2}}\right)\\&=\frac{d}{d\omega}b\left(b^2+(a^2-2b)\omega^2+\omega^4\right)^{-1/2}\\&=\frac{d}{d\omega}bf(\omega)^{-1/2}\\&=-\frac{b}{2}f'(\omega)f(\omega)^{-3/2}\end{aligned}$$

This equals 0 if and only if $f'(\omega) = 4\omega^3 + 2(a^2 - 2b)\omega = 0$ i.e.

$$\omega = \sqrt{\frac{2b - a^2}{2}} = 117.27$$

This is the resonant frequency of the diabetic $\omega_r = \omega_n\sqrt{1 - 2\zeta^2} = 117.27$.

4. From the transfer function we know

$$G(j\omega) = H(j\omega)U(j\omega)$$

i.e.

$$(j\omega)^2 G(j\omega) + bj\omega G(j\omega) + aG(j\omega) = bU(j\omega)$$

As the Fourier transform is linear and using the derivative property we have

$$\ddot{g}(t) + b\dot{g}(t) + ag(t) = bu(t)$$

i.e.

$$\ddot{g}(t) + 36\dot{g}(t) + \left(14.4 \times 10^3\right) g(t) = \left(14.4 \times 10^3\right) u(t)$$

To find the optimal frequency from time domain equation we can assume the input $u(t) = Acos(\omega_0 t)$ and solve the differential equation for g as a function of ω_0. We then maximise the amplitude of the response of g (which will be a sinusoidal) to find the optimal ω_0. It is much easier using the Fourier transform!

5. The transfer function for a series RLC circuit with input $u(t)$ being the voltage of the supply and the output being the voltage across the capacitor is

$$P(j\omega) = \frac{1/(LC)}{(j\omega)^2 + j\omega(L/C) + 1/(LC)}$$

Matching coefficients we have

$$1/(LC) = 14.4 \times 10^3 \text{ and } R/L = 36$$

which implies $R = 36L$ and $C = 1/(14400L)$. Selecting any L will give a valid resistor and capacitor value e.g. $L = 1\,\text{H}$ gives $C = 69.4\,\mu\text{F}$ and $R = 36\,\Omega$.

4: Chernobyl RBMK Reactor: Thermal Feedback and Stability

The stability of a nuclear reactor is affected by *thermal feedback*. As the reactor temperature changes, the reactivity of the reactor can change, which in turn affects the reactor power. A simplified non-linear model for the thermal power $P(t)$ and core temperature $T(t)$ of a nuclear reactor with control-rod temperature, $u(t)$, is:

$$\frac{dP}{dt} = \left(\rho_0 - \beta\left(T(t) - T_r - u(t)\right)\right) P(t) \text{ and } C\frac{dT}{dt} = P - h\left(T(t) - T_C\right) \tag{2}$$

which we linearise about the equilibrium, when $\dot{T} = \dot{P} = 0$, to give the linear model:

$$\frac{dP}{dt} = -\beta P_0\left(T - T_0 - u(t)\right) \text{ and } C\frac{dT}{dt} = P - hT + hT_0 - P_0 = P - h(T - T_C) \tag{3}$$

where

$$T_0 = \rho_0/\beta + T_r \text{ and } P_0 = h\,(T_0 - T_C)$$

T_C is the coolant temperature, $T_r = 300\,\text{C}^\circ$ is a reference temperature, $\rho_0 = 0.01\,\text{s}^{-1}$ is the neutron production at the reference temperature, $\beta = 1 \times 10^4\,(\text{C}^\circ\text{s})^{-1}$ is the thermal feedback rate, $C = 10\,\text{GJ/C}^\circ$ is the thermal capacitance and $h = 1\,\text{MW/C}^\circ$ is the heat feedback.

In the non-linear and linearised models, (2) and (3), the neutron flux $N \propto P$ and the reactivity of and flux from the control rods is assumed to be constant. Typically, the temperature of the coolant is fixed by the ambient temperature of the reservoir used to provide the coolant. We take $T_C = 290\,\text{C}^\circ$. Finally, we note that the input $u(t)$ which is the temperature of the control rods is related to the reactivity of the control rods $v(t)$ by $u(t) = v(t)/\beta$.

1. Define $p(t) = P(t) - P_0$ and $s(t) = T(t) - T_0$. Show that (3) is equivalent to

$$\frac{dp}{dt} = -\beta P_0 s(t) + \beta P_0 u(t) \text{ and } C\frac{ds}{dt} = p(t) - hs(t) \tag{4}$$

   By eliminating $p(t)$ show that the transfer function of the system is given by

$$G(s) = \frac{\beta P_0/C}{s^2 + (h/C)s + \beta P_0/C}$$

2. Find the poles for the system (4) and give conditions under which the system is stable or unstable.

3. Assuming the system is stable show (4) has natural frequency ω_n and damping ratio ζ

$$\omega_n = \sqrt{\frac{\beta P_0}{C}} \text{ and } \zeta = \left(\frac{h}{2}\right)\sqrt{\frac{1}{\beta P_0 C}}$$

4. Find the temperature of the coolant such that the system is critically damped. What happens to the damping ratio ζ as T_C increases if the system is stable?

5. Under certain conditions the RBMK reactor at Chernobyl had a negative thermal feedback rate i.e. $\beta < 0$. Under these conditions what happens to the poles of (4)? What would happen if there was a small positive deviation from the equilibrium temperature T_0?

6. **[Extension]** Plot $T(t)$ for $\beta = -1 \times 10^4\ (\text{C}^\circ\text{s})^{-1}$, $0\ (\text{C}^\circ\text{s})^{-1}$ and $1 \times 10^4\ (\text{C}^\circ\text{s})^{-1}$.

7. **[Extension]** If the nuclear reactor were a mass-spring-damper system what would be the mass (m), spring constant (k) and coefficient of friction (b)?

8. **[Extension]** Show that (3) is linearisation of (2).

> 1. Substituting $p(t) = P(t) - P_0$ and $s(t) = T(t) - T_0$ into (3) we have
>
> $$\frac{dp}{dt} = -\beta P_0 s(t) + \beta P_0 u(t) \text{ and } C\frac{ds}{dt} = p - hs$$
>
> as
>
> $$\frac{ds}{dt} = \frac{d}{dt}(T - T_0) = \frac{dT}{dt}$$

and

$$p(t) - hs(t) = P - P_0 - h(T - T_0)$$

Taking the derivative of $C\dot{s} = p(t) - hs(t)$ and rearranging we have

$$C\ddot{s} + h\dot{s} - \dot{p} = 0$$

Substituting in that $\dot{p} = -\beta P_0 s(t)$ we have

$$C\ddot{s} + h\dot{s} + \beta P_0 s = \beta P_0 u(t)$$

Taking the Laplace transform we have

$$G(s) = \frac{\beta P_0/C}{s^2 + (h/C)s + \beta P_0/C}$$

2. The denominator of the transfer function $G(s)$ is $s^2 + (h/C)s + \beta P_0/C$ so the system is stable so long as $h/C > 0$ and $\beta P_0/C > 0$. Assuming positive thermal and heat feedback $\beta > 0$ and $h > 0$ and positive thermal capacitance $C > 0$, the system is stable only if $P_0 > 0$. This is true only if $T_0 > T_C$. The poles are the roots of

$$s^2 + (h/C)s + \beta P_0/C$$

i.e.

$$s = \frac{-h/C \pm \sqrt{(h/C)^2 + 4\beta P_0/C}}{2} = -5 \times 10^{-5} \pm j3.16$$

Using the values specified for the constants.

3. Matching coefficients with the standard form for a second order transfer function we have

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{\beta P_0/C}{s^2 + (h/C)s + \beta P_0/C}$$

So

$$\omega_n^2 = \frac{\beta P_0}{C} \text{ and } 2\zeta\omega_n = \frac{h}{C}$$

This implies that

$$\omega_n = \sqrt{\frac{\beta P_0}{C}} \text{ and } \zeta = \frac{h}{2}\sqrt{\frac{1}{\beta P_0 C}}$$

4. For the system to be critically damped we require $\zeta = 1$ which holds if and only if

$$\frac{h^2}{4\beta C} = P_0 = h(T_0 - T_C)$$

i.e.

$$T_C = T_0 - \frac{h}{4\beta C} = \frac{\rho_0}{\beta} + T_r - \frac{h}{4\beta C} = 300.0000009975 \sim 300$$

5. For the system to be stable we require $T_C < T_0$ when β, h and $C > 0$. So as T_C increases P_0 decreases which implies that the damping ratio ζ increases i.e. as the temperature of the coolant increases the system becomes more damped. When $T_C = T_0$ the system has a pole at the origin and becomes marginally stable. For $T_C > T_0$ the system is unstable.